

Integration by Parts: When we have a product of functions, we use the formula: $\int u \, dv = uv - \int v \, du$. Choose u to be the function that gets simpler when differentiated. Choose dv to be the rest. Example: $\int x e^x \, dx$. Let $u=x$, $dv=e^x \, dx$. Then $du=dx$, $v=e^x$. Result: $x e^x - e^x + C$.